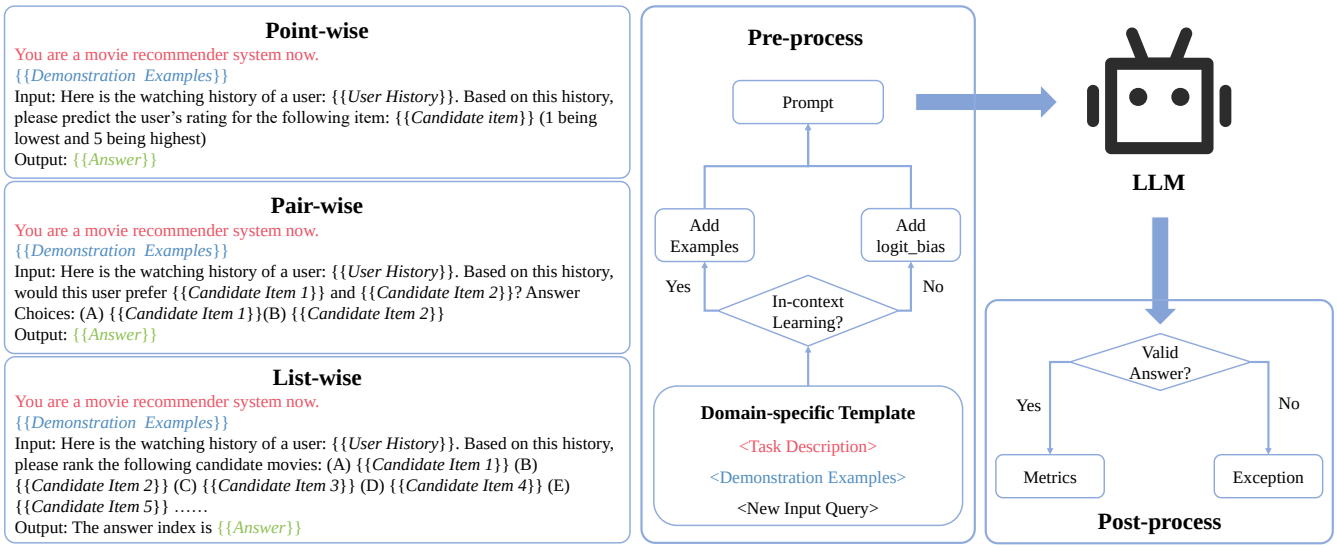

Figure 1: The overall evaluation framework of LLMs for recommendation. The left part demonstrates examples of how prompts are constructed to elicit each of the three ranking capabilities. The right part outlines the process of employing LLMs to perform different ranking tasks and conduct evaluations.

have commonly utilized three approaches to construct these ranking lists: point-wise, pair-wise, and list-wise [18, 42]. Consequently, in this paper, our specific focus is on probing the recommendation capabilities of LLMs by aligning them with these three ranking perspectives. Detailed formulation of these three ranking perspectives could be seen in Section 3.

To investigate the potential of LLMs in recommendation tasks from these three ranking perspectives, we begin by reformulating the three capabilities into prompts that are tailored to the specific domain and serve as input for LLMs. We then conduct an empirical analysis of ChatGPT and other popular LLMs from OpenAI on four widely-used recommendation benchmarks from different knowledge-rich domains. To the best of our knowledge, this is the first empirical study to probe the capabilities of ChatGPT in recommender systems from different ranking perspectives.

Major Findings. In summary, we have the following major findings after empirical experiments:

- ChatGPT shows consistent advantages in all three ranking capabilities compared with other LLMs.
- ChatGPT is good at list-wise and pair-wise ranking while less good at point-wise ranking.
- ChatGPT can outperform traditional recommendation models with limited training data.
- Considering the improvements with cost, we recommend list-wise ranking for LLM-based recommenders in practice.
- ChatGPT exhibits potential in explainable recommendations and a good understanding of item similarity.

We hope that this preliminary evaluation of ChatGPT in recommendation can provide new perspectives on both assessing the capabilities of LLMs and utilizing LLMs, such as ChatGPT, to enhance recommender systems.

2 BACKGROUND

2.1 Large Language Models

Pioneering studies [3, 27] demonstrated that LLMs can perform a diverse range of tasks without requiring gradient updates, solely based on textual instructions or a few examples. This has drawn significant attention towards improving the capabilities of LLMs. Previous studies [15] have investigated the performance limits of pre-trained language models (PLMs) by training larger models, as they have noted that augmenting the model or data size typically enhances the model's ability on downstream tasks, such as Megatron-turing NLG [35] with 530B parameters, Gopher [28] with 280B parameters, Ernie 3.0 Titan [39] with 260B parameters, BLOOM [33] with 175B parameters, and PaLM [5] with 540B parameters. These LLMs have exhibited exceptional performance on challenging tasks, showcasing new abilities that were not apparent in smaller pre-trained language models. For a more comprehensive overview of LLMs, we would recommend referring to [44].

2.2 Language Models for Recommendation

The remarkable success of pre-trained LMs in NLP community has motivated researchers in recommender systems to explore their potential in recommendation tasks. Existing works can be categorized into two types: (i) utilizing LMs training strategies to reformulate and model recommendation tasks, such as BERT4Rec (*masked language modeling*) [36], UnisRec (*pre-train and finetune*)[13], P5 (*pre-train and prompting*) [9] and (ii) using LMs to obtain better representations of users and items as extra features based on textual information [40]. More recently, some researchers have explored leveraging off-the-shelf pre-trained LMs as recommender systems by reformulating the recommendation tasks with prompts as multi-token cloze tasks [25, 34, 43]. In this paper, we aim to conduct a